

5. A group of students were asked to calculate the population size of garden snails (*Helix aspersa*) in 5 different areas of similar size in the school grounds. They sampled the five areas of the school grounds early in the morning on a damp day. The students sampled each area once only and counted the number of snails they found. The shell of each snail was marked with a small spot of white ink. The snails were then released.

Garden snail marked with white ink



Approximately one week later, on a damp morning, the students sampled the same areas of the school grounds a second time. This time they recorded the number of recaptured snails (the ones that had been marked with white ink a week earlier) and the number of snails which had not been marked with white ink.



The table below shows the capture – recapture results obtained by the students.

Area of school grounds	No. of snails captured and marked in 1 st sample	No. of snails in 2 nd sample	No. of snails in 2 nd sample previously marked	Population size
playing field	3	2	1	6
hedgerow	9	4	3	12
flowerbed	7	4	2	14
boundary wall	11	9	5	20
vegetable garden	23	17	10	

- (a) Use the equation below to calculate the population size for the snails living in the vegetable garden. **Write your answer in the table above.** [2]

population size = $\frac{\text{number in 1}^{\text{st}} \text{ sample} \times \text{number in 2}^{\text{nd}} \text{ sample}}{\text{number in 2}^{\text{nd}} \text{ sample previously marked}}$

Space for working

- (b) State **one** way in which the students kept their investigation fair. [1]

.....

3430UD01
11



Examiner
only

- (c) The song thrush (*Turdus philomelos*) feeds on snails. During the week after marking the snails with white ink some of the students noticed that the number of song thrushes in the five areas of the school grounds had increased. This increase may have been due to a problem resulting from the students' method. Suggest why the number of song thrushes increased and how the students could overcome the problem. [2]

Song thrush eating garden snail



.....

.....

.....

.....

5

